



Assignment: Multi-Tenant Feature Flag Management System

Overview

Build a small **SaaS - like feature-flag management system** with **three separate front-end applications** and a **Node.js backend**. The front-end doesn't have to look polished and professional. Basic UI usability is enough. Candidates can use AI, but should use it as a mere accelerator to execute your ideas into code. If the AI ideates, we believe your ability to come up with innovations would be stifled.

The system allows:

- A **Super Admin** to create organizations
- **Organization Admins** to manage feature flags
- **End Users** to check whether specific features are enabled for their organization

This assignment is designed to evaluate your ability to design APIs, model data, manage roles, and make practical engineering trade-offs.

Tech Constraints

- **Backend:** Node.js with any framework (eg. Express)
 - **Frontend:** Any framework or plain HTML/JS
 - **Database:** Any (SQL or NoSQL)
 - **Authentication:** Custom implementation only, No third-party auth providers (Auth0, Firebase Auth, Cognito, etc.)
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System Roles

1. Super Admin

- Uses **static credentials** (hardcoded or config-based) that can:
 - Log in
 - Create organizations
 - View list of organizations

2. Organization Admin

- Belongs to one organization and can:
 - Sign up
 - Log in
 - Create, update, and delete feature flags for their organization

3. End User

- Belongs to one organization and can access a simple form / checkbox to submit and check whether a feature is enabled for their organization
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Applications

1. Super - Admin Frontend

Used by **Software Host**

Required features

- Login
 - Create Organizations
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2. Admin Frontend

Used by **Organization Admins.**

Required features

- Signup
 - Login
 - Feature flag management:
 - Create a feature flag (e.g., `feature_key`)
 - Enable or disable the feature
 - Feature flags are scoped to the admin's organization
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3. User Frontend

Used by **End Users.**

Required features

- Simple page with:
 - A checkbox or input for a feature key
 - A submit button
 - On submission, the system should respond whether:
 - The feature is **enabled or disabled** for the user's organization
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Data Requirements

We're expecting persistent storage of:

- Organisations
- Users
- Roles
- Feature flags

Time Guidance

This task is expected to take **6–10 hours**. It is **not** expected to be production-ready on submission. We focus more on the clarity, and decision making on the work being submitted rather than treating the code like a Pull Request review.